RAG Learning

Edexcel GCSE Geography B (1GB0) — Component 2: UK Geographical Issues · Topic 6: Geographical Investigations

2.4 Geographical Investigations

1. Geographical Investigations in Edexcel B

Fieldwork is examined in **Component 2, Section C (Topic 6: Geographical Investigations)**. You must complete **two investigations**, each in a different environment:

Investigation	Choose one of
Physical	Coasts (e.g. how coastal management affects beach profiles and longshore drift) **or** Rivers (e.g. how channel characteristics change downstream)
Human	Dynamic urban areas (e.g. how and why quality of life varies within a city) **or** Changing rural settlements (e.g. how and why rural services and populations are changing)

- Questions test both your own fieldwork and **unfamiliar** fieldwork contexts, using data and methods you have not seen before.
- You should be able to use both **quantitative** data (numbers — measurements, counts) and **qualitative** data (descriptions, photographs, interviews).
- The final question in each investigation section is usually an 8-mark evaluation of your methods, results or conclusions.

2. The Six Stages of a Geographical Enquiry

Stage	What you need to know
1. Suitable question	Formulate an enquiry question or hypothesis that can be investigated through fieldwork. Know the geographical theory or concept behind it (e.g. the Bradshaw

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	model for rivers) and why the location was suitable (safe, accessible, shows the feature being studied).
2. Selecting, measuring and recording data	Primary data (collected yourself) and secondary data (collected by others, e.g. census data, Environment Agency flood data, maps). Sampling methods: **random**, **systematic** (at regular intervals) and **stratified** (in proportion to sub-groups). Know the equipment used and how to reduce risks.
3. Processing and presenting data	Choose suitable methods — line graphs, bar charts, pie charts, scatter graphs, cross-sections, choropleth maps, annotated photographs, flow-line maps. Justify why each was appropriate.
4. Describing, analysing and explaining	Describe patterns and trends, identify anomalies, use statistics (mean, median, mode, range, interquartile range) and explain the results using geographical theory.
5. Reaching conclusions	Answer the original question or say whether the hypothesis was proved, supported by evidence.
6. Evaluation	Identify problems with data collection (accuracy, reliability, sample size, weather, human error), limitations of the methods and conclusions, and suggest improvements and extensions.

3. Example Physical Enquiry: River Characteristics Downstream

Hypothesis: "Channel width, depth and velocity increase with distance downstream."

- **Theory:** the **Bradshaw model** predicts that width, depth, discharge and velocity increase downstream, while gradient and load particle size decrease.
- **Method:** systematic sampling at several sites along the river. Width measured with a tape measure; depth with a metre rule at regular intervals across the channel; velocity by timing a float over 10 m (repeated three times and averaged); pebble size with a ruler or callipers (10 pebbles per site, chosen randomly).
- **Presentation:** cross-sections for each site; scatter graphs of distance downstream against width, depth and velocity, with a line of best fit.

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- **Risks:** drowning, slipping, cold water, waterborne disease (Weil's disease) — reduced by wearing wellingtons, working in groups, avoiding high-flow conditions and washing hands.
- **Evaluation:** only a few sites were sampled; a single visit means results reflect that day's flow; floats can get stuck; human error reading measurements. Improvements: more sites, using a flow meter, repeating on several dates.

4. Example Human Enquiry: Quality of Life Across a City

Question: "How and why does the quality of the environment vary from the inner city to the suburbs?"

- **Theory:** urban land-use models (e.g. the Burgess concentric zone model) and the idea of inequality within cities.
- **Method:** a transect from the city centre outwards with systematic sampling every 500 m. At each site: an environmental quality survey (bipolar scoring of litter, graffiti, green space, building condition), pedestrian or traffic counts, noise readings, questionnaires using random or stratified sampling. Secondary data: Index of Multiple Deprivation, census data and house prices.
- **Presentation:** line graphs of score against distance; choropleth maps of deprivation; radar charts comparing sites; annotated photographs.
- **Evaluation:** environmental quality surveys are subjective (different students score differently); a small number of questionnaires may not be representative; time of day affects pedestrian counts. Improvements: group scoring with agreed criteria, larger samples and repeat visits.

5. Geographical Skills

Skills can be tested in all three papers. Make sure you can:

- **Cartographic:** use four- and six-figure grid references, scale and distance, compass directions, contour lines and spot heights on OS maps; describe and explain patterns on atlas and choropleth maps; interpret cross-sections and transects.

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- **Graphical:** construct and interpret line graphs, bar charts, pie charts, scatter graphs (with lines of best fit), population pyramids, climate graphs, triangular graphs and dispersion diagrams.
- **Numerical and statistical:** calculate percentages, percentage change, mean, median, mode, range, quartiles and interquartile range; understand correlation and the difference between correlation and causation.
- **Formulating enquiry:** identify questions, sequence investigations and evaluate sources.
- **Literacy:** write extended answers using specialist terms accurately — SPaG marks are awarded on selected questions in every paper.

Exam Tip: In fieldwork questions on your own enquiry, always name your location and give specific details (e.g. "we measured at 6 sites along Carding Mill Valley, 200 m apart"). Generic answers that could apply to anyone's fieldwork are capped at the lower levels.

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